

Multiple Choice

1. **Answer:** A

Options: A) We don't know for sure.; B) aliens (panspermia).; C) comets.; D) the Earth's oceans.

Note: Correct option: A - We don't know for sure.

2. **Answer:** C

Options: A) The velocity is the same on every point on the disk; B) The part closer to the axis has a longer period; C) The period of rotation is the same on every point on the disk; D) The part closer to the axis has a shorter period

Note: Correct option: C - The period of rotation is the same on every point on the disk

3. **Answer:** C

Options: A) $m \cdot v$; B) $0.5 \cdot m / (R^3)$; C) $0.5 \cdot m \cdot (2 \text{ GM}/R)$; D) $0.6 \cdot G(M^2)/R$

Note: Correct option: C - $0.5 \cdot m \cdot (2 \text{ GM}/R)$

4. **Answer:** D

Options: A) 9.3×10^{13} meters; B) 9.3×10^{10} meters; C) 1.5×10^{14} meters; D) 1.5×10^{17} meters

Note: Correct option: D - 1.5×10^{17} meters

5. **Answer:** C

Options: A) They reveal that the early solar system consisted mostly of hydrogen and helium gas.; B) They reveal that meteorites are much older than the comets and planets.; C) They reveal that the age of the solar system is approximately 4.6 billion years.; D) They reveal that the solar system once contained 10 planets.

Note: Correct option: C - They reveal that the age of the solar system is approximately 4.6 billion years.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It's too diffuse'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'A and B'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'it switches polarity every 11 years'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** The spectra of some meteorites are similar to the spectra of asteroids in the asteroid belt.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Amino acids with a preferred orientation or "chirality". Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key